Question I

Consider the following gradient flow with discrete normalization (GFDN)

$$\begin{aligned}
&\partial_t u(x,t) = \left[\partial_{xx} - V(x) - \beta|u|^2\right] u, \quad 0 < x < 1,\ t_n \le t < t_{n+1},\ n \ge 0,\\
&u(x,t_{n+1}) = u(x,t_{n+1}^+) := \frac{u(x,t_{n+1}^-)}{\|u(\cdot,t_{n+1}^-)\|_{L^2}}, \qquad 0 \le x \le 1, \quad n \ge 0,\\
&u(x,0) = g_0(x), \qquad 0 \le x \le 1,\\
&u(0,t) = u(1,t) = 0, \qquad t \ge 0,
\end{aligned}$$

where $u = u(x,t)$ is a real-valued function, $V(x) \ge 0$ for $0 \le x \le 1$ is a given function, $\beta \ge 0$ is a given constant, $t_n = n\tau$ for $n = 0, 1, 2, \ldots$ with $\tau > 0$ being the time step, $u(x,t_n^\pm) := \lim_{t\to t_n^\pm} u(x,t)$, $\|u\|_{L^2}^2 = \int_0^1 |u(x)|^2\,dx$ and $g_0(x)$ is a given function satisfying $\|g_0\|_{L^2} = 1$. Define the mass and energy as

$$\begin{aligned}
M(t) &:= M(u(\cdot,t)) = \int_0^1 |u(x,t)|^2\,dx, \qquad t \ge 0,\\
E(t) &:= E(u(\cdot,t)) = \int_0^1 \left[|\partial_x u(x,t)|^2 + V(x)|u(x,t)|^2 + \frac{\beta}{2}|u(x,t)|^4\right] dx.
\end{aligned}$$

1. Show that the mass and energy are diminishing in each time interval $[t_n, t_{n+1})$, i.e.

$$M(t_2) \le M(t_1), \qquad E(t_2) \le E(t_1), \qquad t_n \le t_1 \le t_2 < t_{n+1}, \quad n \ge 0.$$

2. When $\beta = 0$, show that

$$E(g_0) = E(u(x,t_0)) \ge E(u(x,t_1)) \ge \cdots \ge E(u(x,t_n)) \ge \cdots, \qquad n \ge 0,$$

for any given time step $\tau > 0$ and initial data $g_0(x)$.

3. Let $\tau \to 0^+$ in the problem GFDN, what partial differential equations can you get? Show that the limiting equation is mass conservative and energy diminishing.

Question II

Define a dynamical system

$$\frac{dy_t}{dt} = -A^T(Ax_t - b), \tag{1a}$$

$$y_t \in \partial\psi(x_t), \tag{1b}$$

where $A \in \mathbb{R}^{n\times k}$ satisfies that A^TA has smallest eigenvalue $\gamma > 0$, $b = Ax^*$, and $\psi(x) = \|x\|_1 + \frac{\|x\|_2^2}{2\alpha}$ has its subgradient set $\partial\psi(x)$ at $x \in \mathbb{R}^k$, where $\alpha > 0$ is a given constant. For a convex function $\psi : \mathbb{R}^k \to \mathbb{R}$, define the *Bregman divergence* function associated with ψ, $D_\psi : \mathbb{R}^k \times \mathbb{R}^k \to \mathbb{R}$, by

$$D_\psi(x', x) := \psi(x') - \psi(x) - \langle \partial\psi(x), x' - x\rangle. \tag{2}$$

1. Show that (1) leads to the following ODE

$$\frac{dD_\psi(x^*, x_t)}{dt} = -\|A(x_t - x^*)\|_2^2. \tag{3}$$

2. For x_t in (1), define

$$\tau := \inf\{t > 0 : \text{sign}(x_t) = \text{sign}(x^*)\}. \tag{4}$$

Denote by $\underline{x}^*$ the smallest nonzero magnitude of $|x_i|$ $(i = 1, \ldots, k)$. Find an upper bound of τ, as tight as possible.